Week 9 – Advanced Conditionals & Loops

■ Learning Objectives

• Use nested and combined conditionals effectively.
• Understand break and continue statements in depth.
• Control loop flow with multiple conditions and flags.
• Apply these skills in practical problem-solving scenarios.

■ Key Concepts

1. Nested If: Allows multiple condition layers inside each other.
2. Logical Operators: Combine conditions with and/or/not.
3. Break & Continue: Exit or skip loop iterations strategically.
4. Sentinel Loops: Repeat until a stop condition is met.

Example 1 – Nested If Statements

```
x = int(input("Enter a number: "))
if x > 0:
    if x % 2 == 0:
        print("Positive even number")
    else:
        print("Positive odd number")
else:
    if x < 0:
        print("Negative number")
    else:
        print("Zero")
```

Example 2 – Combined Conditions

```
age = int(input("Enter your age: "))
if 13 <= age <= 19:
    print("Teenager")
elif age < 13 or age > 19:
    print("Not a teenager")
```

Example 3 – Break and Continue

```
for i in range(1, 10):
    if i == 5:
        continue  # Skip 5
    if i == 8:
        break     # Stop at 8
    print(i)
```

Example 4 – While Loop with Condition Flags

```
found = False
nums = [1, 3, 5, 7, 9]
target = 7

i = 0
while i < len(nums) and not found:
    if nums[i] == target:
        found = True
    else:
        i += 1

print("Found:", found)
```

■ Practice Problems

Practice 1 – Number Guessing Game

```
import random
secret = random.randint(1, 20)
while True:
    guess = int(input("Guess a number (1–20): "))
    if guess == secret:
        print("You got it!")
        break
    elif guess < secret:
        print("Too low!")
    else:
        print("Too high!")
```

Practice 2 – Count Evens and Odds

```
evens = odds = 0
while True:
    n = int(input("Enter number (0 to stop): "))
    if n == 0:
        break
    if n % 2 == 0:
        evens += 1
    else:
        odds += 1
print("Evens:", evens, "Odds:", odds)
```

Practice 3 – Skip Multiples of 3

```
for i in range(1, 21):
    if i % 3 == 0:
        continue
    print(i, end=" ")
```

■ Homework Challenge

Create a program that asks the user for a series of numbers and prints the total sum when the user types 'done'.

```
total = 0
while True:
    s = input("Enter a number or 'done': ")
    if s.lower() == 'done':
        break
    if s.isdigit():
        total += int(s)
print("Total =", total)
```